Example 3.26 Solve $2x^2 - 2\sqrt{6}\,x + 3 = 0$

Solution $2x^2 - 2\sqrt{6}\,x + 3 = 2x^2 - \sqrt{6}\,x - \sqrt{6}x + 3$ (by spliting the middle term)

$$= \sqrt{2}x\left(\sqrt{2}x - \sqrt{3}\right) - \sqrt{3}\left(\sqrt{2}x - \sqrt{3}\right) = \left(\sqrt{2}x - \sqrt{3}\right)\left(\sqrt{2}x - \sqrt{3}\right)$$

Now, equating the factors to zero we get,

$$\left(\sqrt{2}x - \sqrt{3}\right)\left(\sqrt{2}x - \sqrt{3}\right) = 0$$

$$\left(\sqrt{2}x - \sqrt{3}\right)^2 = 0$$

$$\sqrt{2}x - \sqrt{3} = 0$$

$\therefore$ the solution is $x = \dfrac{\sqrt{3}}{\sqrt{2}}$.

Example 3.27 Solve $2m^2 + 19m + 30 = 0$

Solution $2m^2 + 19m + 30 = 2m^2 + 4m + 15m + 30 = 2m(m+2) + 15(m+2)$

$$= (m+2)(2m+15)$$

Equating the factors to zero we get,

$(m+2)(2m+15) = 0$

$m + 2 = 0 \Rightarrow m = -2$ or $2m + 15 = 0$ we get, $m = \dfrac{-15}{2}$

Therefore, the roots are $-2, \dfrac{-15}{2}$.

Some equations which are not quadratic can be solved by reducing them to quadratic equations by suitable substitutions. Such examples are illustrated below.

Example 3.28 Solve $x^4 - 13x^2 + 42 = 0$

Solution Let $x^2 = a$. Then, $(x^2)^2 - 13x^2 + 42 = a^2 - 13a + 42 = (a-7)(a-6)$

Given, $(a-7)(a-6) = 0$ we get, $a = 7$ or 6.

Since $a = x^2$, $x^2 = 7$ then, $x = \pm\sqrt{7}$ or $x^2 = 6$ we get, $x = \pm\sqrt{6}$

Therefore, the roots are $x = \pm\sqrt{7}, \pm\sqrt{6}$

Example 3.29 Solve $\dfrac{x}{x-1} + \dfrac{x-1}{x} = 2\dfrac{1}{2}$

Solution Let $y = \dfrac{x}{x-1}$ then $\dfrac{1}{y} = \dfrac{x-1}{x}$.

Therefore, $\dfrac{x}{x-1} + \dfrac{x-1}{x} = 2\dfrac{1}{2}$ becomes $y + \dfrac{1}{y} = \dfrac{5}{2}$

$2y^2 - 5y + 2 = 0$ then, $y = \dfrac{1}{2}, 2$

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